

```
- require:
- group: {{ sls }}.group
```

Now let us set up the webserver configuration file *config.sls* as below:

```
[root@ip-172-31-31-154 kitchen-salt]# cat webserver/config.sls
{% from "webserver/map.jinja" import vars with context %}
include:
- webserver.install
{{ sls }}.index.html:
file.managed:
- name: {{ pillar['doc_root'] }}/index.html
- source: salt://webserver/files/index.html
- require:
- sls: webserver.install
{{ sls }}.server.conf:
file.managed:
- name: {{ pillar['server_conf'] }}
- source: salt://webserver/templates/httpd.conf.j2
- template: jinja
- require:
- sls: webserver.install
```

Next, configure the *service.sls* to start webserver service once setup is completed.

```
[root@ip-172-31-31-154 kitchen-salt]# cat webserver/service.sls
{% from "webserver/map.jinja" import vars with context %}
include:
- webserver.install
- webserver.config
{{ sls }}.service:
service.running:
- name: {{ vars['service'] }}
- watch:
- sls: webserver.install
- sls: webserver.config
```

Installing Kitchen

Once the formula is ready, let's install the Kitchen-specific dependencies using Gemfile, which requires *kitchen-salt*, *kitchen-docker*, and *kitchen-sync* libraries.

```
[root@ip-172-31-31-154 kitchen-salt]# cat Gemfile
#Gemfile
source 'https://rubygems.org'
gem 'kitchen-salt'
gem 'kitchen-docker'
gem 'kitchen-sync'
```

Next, run the installer which installs all the required dependencies using Gem bundler:

```
[root@ip-172-31-31-154 kitchen-salt]# gem install bundler
Fetching bundler-0.0.1.gem
Successfully installed bundler-0.0.1
Parsing documentation for bundler-0.0.1
Installing ri documentation for bundler-0.0.1
Done installing documentation for bundler after 0 seconds
1 gem installed
[root@ip-172-31-31-154 kitchen-salt]# bundler install | grep
kitchen
Using test-kitchen 3.3.2
Using kitchen-docker 2.13.0
Using kitchen-salt 0.7.2
Using kitchen-sync 2.2.1
```

Configuring Kitchen

Once the Kitchen dependencies are sorted, let's create a *.kitchen.yml* file consisting of Test Kitchen configurations. This is the main file responsible for spinning up a Docker container and deploying the Salt formula.

The file includes the following major components.

- **driver:** Defines which driver needs to be configured and how to control it. In our example, we will use Docker as a default driver.
- **platforms:** Defines the platforms or distributions against which it tests Salt states or formulas. We will use CentOS7 as a default platform for this demonstration.
- **provisioner:** Defines what plugins to use to converge the system and how to apply the changes. The provisioner we use is salt-solo.
- **suites:** Defines the software suite to be used to test Kitchen against. In this context, Kitchen will test against 3004.2 release of Salt.
- **verifier:** Defines what tests need to be performed against the newly provisioned infrastructure. We will use TestInfra as a default testing framework to run our tests.

The next step is to put all the configs together:

```
[root@ip-172-31-31-154 kitchen-salt]# cat .kitchen.yml
driver:
  name: docker
  use_sudo: false
  privileged: true
  run_command: /usr/sbin/init
  forward:
    - 80
provisioner:
  name: salt_solo
  salt_install: bootstrap
```